

Logic Continued

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Atomic Formula

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We can combine atomic formulas using $\vee, \wedge, \neg$ to build larger formulas.

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Interpretation

An **interpretation** $\mathcal{A}$ in prop. logic is **suitable** for a formula F if it assigns each atomic formula in F a truth value.

Interpretation Example

Take the formula

$$F = (A_1 \vee A_2) \wedge A_3$$

and the interpretation $\mathcal{A}$ over $Z = \{A_1, A_2, A_3\}$ with

$$\mathcal{A}(A_1) = 0, \mathcal{A}(A_2) = 1, \mathcal{A}(A_3) = 1$$

$\mathcal{A}$ is suitable for F . And it is even a **model**, since the formula evaluates to true under the truth assignment.

Interpretation Example

Take the formula

$$F = (A_1 \vee A_2) \wedge A_3$$

and the interpretation $\mathcal{B}$ over $Z = \{A_1, A_3\}$ with

$$\mathcal{B}(A_1) = 1, \mathcal{B}(A_3) = 1$$

$\mathcal{B}$ is **not** suitable for F since it does not assign a value to A_2 .

Note that even though for any truth assignment we could add for A_2 to $\mathcal{B}$, F would evaluate to true, it is still not suitable!

- ① **Variable symbols** are of the form x_i . (We often use x, y, z .)
- ② $f_i^{(k)}$ is a **function** with k arguments. (We often omit the argument count and just write f, g, h .)
- ③ A **constant** is a function with no arguments.
- ④ $P_i^{(k)}$ is a **predicate** symbol with k arguments. (We just write P, Q, R .)
- ⑤ A **term** can be built with variables and functions.

- 1 For some terms $t_1, \dots, t_k$ an **atomic formula** is $P_i^{(k)}(t_1, \dots, t_k)$.
- 2 For formulas F and G , $F \wedge G$, $F \vee G$, $\neg F$ are formulas.
- 3 For a formula F , $\forall x_i F$ and $\exists x_i F$ are formulas

Syntax Examples

x_1 is a term and $f_1(x_1)$ is a term. Now if we have a function $f_2^{(2)}$ we can combine them to

$$f_2(x_1, f_1(x_1)).$$

$P(x_1)$ is a formula and $P(f_1(x_1))$ is too. We can combine them to

$$P(x_1) \wedge P(f_1(x_1)).$$

$P(x_2)$ is a formula too. And so is

$$\forall x_1 P(x_2).$$

Note that there is no reason why x should appear in the formula F for $\forall x F$.

Free Variables

Free Variable

A *free variable* not bound by a quantifier. All free variables have to be defined by an interpretation of the formula.

For example, in the formula $\forall xP(x, y)$, the x is bound by the quantifier, but y is free.

Substitution

$F[x/t]$ is the formula F , but we substitute every free occurrence of x with t .

For example, $P(x)[x/f(y)] = P(f(y))$.

An **interpretation** needs to define:

- 1 The universe U we are working in
- 2 All function symbols.
- 3 All predicate symbols
- 4 All *free* variables

All the functions defined go from U to U and all the values assigned to free variables are in U .

If an interpretation $\mathcal{A}$ defines all this for a formula F , then it is **suitable** for that formula.

We write $U^{\mathcal{A}}, x^{\mathcal{A}}, f^{\mathcal{A}}, P^{\mathcal{A}}$ for the stuff defined by $\mathcal{A}$.

Interpretation Example

Say we have the formula

$$F = \forall x(P(x) \vee P(f(x, y))).$$

A suitable interpretation should define U, P, f, y . Say we take $\mathcal{A}$ with:

$$U^{\mathcal{A}} = \mathbb{N}$$

$$P^{\mathcal{A}}(x) = \text{even}(x)$$

$$f^{\mathcal{A}}(x, y) = x + y$$

$$y^{\mathcal{A}} = 1$$

Then it reads "for all natural numbers x , either x or $x + 1$ is even". And $\mathcal{A}$ is a model for F .

In prop. logic we interpreted the variables $A, B, \dots$ with **truth values**. In predicate logic we interpret the free variables with **values from the universe**. So something like this:

$$P(x_1) \vee x_2$$

is **not** a valid formula.

(We can however achieve something similar by using predicates, for example of the form $P_i^{(0)}$.)

For some formula F and interpretation $\mathcal{A}$ we want to find the truth value $\mathcal{A}(F)$.

- For terms, functions, variables we simply evaluate them as defined
- We also evaluate predicates as defined by $\mathcal{A}$

Overwriting

$\mathcal{A}_{[x \rightarrow u]}$ is the same as $\mathcal{A}$, but we overwrite x with u .

- $\mathcal{A}(\forall x F) = 1$ if $\mathcal{A}_{[x \rightarrow u]}(F) = 1$ for all u in U
- $\mathcal{A}(\exists x F) = 1$ if $\mathcal{A}_{[x \rightarrow u]} = 1$ for some u in U

Useful Properties

You can find some useful properties in Lemma 6.7 in the script. Note that for a lot of exercises you are not allowed to use those (if explicitly stated).

Semantics Example

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Intuition: If $P(x)$ holds for all values in the universe, then there cannot be any value that $f(x)$ maps to, for which $P(x)$ does not hold. So let's do this formally:

(More) Formally

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We show that $\forall x P(x) \models \neg \exists x (\neg P(f(x)))$. Take any interpretation $\mathcal{A}$ with $\mathcal{A}(\forall x P(x)) = 1$.

$$\iff \mathcal{A}_{[x \rightarrow u]} P(x) = 1 \text{ for all } u \text{ in } U \quad (1)$$

(More) Formally

We show that $\forall x P(x) \models \neg \exists x (\neg P(f(x)))$. Take any interpretation $\mathcal{A}$ with $\mathcal{A}(\forall x P(x)) = 1$.

$$\iff \mathcal{A}_{[x \rightarrow u]} P(x) = 1 \text{ for all } u \text{ in } U \quad (1)$$

Now consider $\mathcal{A}(\neg \exists x (\neg P(f(x)))) = 1$

$$\iff \mathcal{A}(\exists x (\neg P(f(x)))) = 0 \quad (2)$$

$$\iff \mathcal{A}_{[x \rightarrow u]} (\neg P(f(x))) = 0 \text{ for all } u \text{ in } U \quad (3)$$

$$\iff \mathcal{A}_{[x \rightarrow u]} P(f(x)) = 1 \text{ for all } u \text{ in } U \quad (4)$$

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$$\iff \mathcal{A}_{[x \rightarrow u]} P(f(x)) = 1 \text{ for all } u \text{ in } U \quad (4)$$

Since $f(x)$ is an element in U and the right side holds we know that the left side also needs to hold.

How to Prove Stuff Like This

Most proofs work the same way:

- 1 Apply definitions until you get to language.
- 2 Reason with language

Note that in the example, we would need to justify all the steps with "(semantics of ...)" on each line!

Exercise 1

Prove or disprove each of the following statements. Do not use any theorems or lemmas from the lecture notes. Note that x may appear free in F , G or both.

a) For any formulas F and G , we have

$$\forall x (F \wedge G) \models (\forall x F) \wedge G.$$

b) For any formulas F and G , we have

$$\exists x (F \wedge G) \models (\exists x F) \wedge G.$$

a) The statement is true.

Proof. Let $\mathcal{A}$ be any interpretation suitable for both $\forall x (F \wedge G)$ and $(\forall x F) \wedge G$, such that $\mathcal{A}(\forall x (F \wedge G)) = 1$. According to the semantics of $\forall$, we have $\mathcal{A}_{[x \rightarrow u]}(F \wedge G) = 1$ for all $u \in U$. According to the semantics of $\wedge$, we further have (1) $\mathcal{A}_{[x \rightarrow u]}(F) = 1$ for all $u \in U$ and (2) $\mathcal{A}_{[x \rightarrow u]}(G) = 1$ for all $u \in U$.

The fact (1) implies (3) $\mathcal{A}(\forall x F) = 1$, according to the semantics of $\forall$. Furthermore, note that if x appears free in G , then it also appears free in $(\forall x F) \wedge G$, and since $\mathcal{A}$ is suitable for $(\forall x F) \wedge G$, it must assign a value to x . We now define u^* as follows: if x appears free in G , then u^* is the value assigned to x by $\mathcal{A}$, else u^* is arbitrary. By the definition of u^* , we have $\mathcal{A}_{[x \rightarrow u^*]}(G) = \mathcal{A}(G)$, so by (2), we have (4) $\mathcal{A}(G) = 1$.

The facts (3) and (4) imply that $\mathcal{A}((\forall x F) \wedge G) = 1$.

b) The statement is false.

Counterexample. Let $F = P(x)$ and $G = Q(x)$. Let $\mathcal{A}$ be the interpretation with the universe $U^{\mathcal{A}} = \{0, 1\}$, which defines:

- $P^{\mathcal{A}}(0) = 1$ and $P^{\mathcal{A}}(1) = 1$
- $Q^{\mathcal{A}}(0) = 1$ and $Q^{\mathcal{A}}(1) = 0$
- $x^{\mathcal{A}} = 1$

We then have $\mathcal{A}(\exists x (P(x) \wedge Q(x))) = 1$, because $\mathcal{A}_{[x \rightarrow 0]}(P(x) \wedge Q(x)) = 1$. However, $\mathcal{A}((\exists x P(x)) \wedge Q(x)) = 0$, because $\mathcal{A}(Q(x)) = 0$.

Exercise 2

Prove the following statements using the semantics of predicate logic (Definition 6.36). Do not use other results from the lecture notes.

a) $\exists x \forall y P(x, y) \models \exists x P(x, f(x)).$

b) $\neg(\forall x P(x)) \models \exists x \neg P(x).$

- a) Let $\mathcal{A}$ be an interpretation which is suitable both for $\exists x \forall y P(x, y)$ and for $\exists x P(x, f(x))$, and such that $\mathcal{A}$ is a model for $\exists x \forall y P(x, y)$, meaning $\mathcal{A}(\exists x \forall y P(x, y)) = 1$. In particular, observe that $f^{\mathcal{A}} : U^{\mathcal{A}} \rightarrow U^{\mathcal{A}}$ is defined. We have

$$\mathcal{A}(\exists x \forall y P(x, y)) = 1$$

$$\begin{aligned} &\dot{\iff} \mathcal{A}_{[x \rightarrow u]}(\forall y P(x, y)) = 1 \text{ for some } u \in U^{\mathcal{A}} && \text{(Semantics of } \exists) \\ &\dot{\iff} \mathcal{A}_{[x \rightarrow u][y \rightarrow v]}(P(x, y)) = 1 \text{ for some } u \in U^{\mathcal{A}} \text{ and all } v \in U^{\mathcal{A}} && \text{(Semantics of } \forall) \\ &\implies \mathcal{A}_{[x \rightarrow u][y \rightarrow f^{\mathcal{A}}(u)]}(P(x, y)) = 1 \text{ for some } u \in U^{\mathcal{A}} && (f^{\mathcal{A}}(u) \in U^{\mathcal{A}}) \\ &\dot{\iff} \mathcal{A}_{[x \rightarrow u]}(P(x, f(x))) = 1 \text{ for some } u \in U^{\mathcal{A}} && \text{(Semantics of } f(x)) \\ &\dot{\iff} \mathcal{A}(\exists x P(x, f(x))) = 1. && \text{(Semantics of } \exists) \end{aligned} \tag{1}$$

- b) Let $\mathcal{A}$ be an interpretation which is suitable both for $\neg(\forall x P(x))$ and for $\exists x \neg P(x)$, and such that $\mathcal{A}$ is a model for $\neg(\forall x P(x))$, meaning $\mathcal{A}(\neg(\forall x P(x))) = 1$. We have

$$\begin{aligned}
 & \mathcal{A}(\neg(\forall x P(x))) = 1 \\
 & \iff \mathcal{A}(\forall x P(x)) = 0 && \text{(Semantics of } \neg) \\
 & \iff \mathcal{A}_{[x \rightarrow u]}(P(x)) = 0 \text{ for some } u \in U^{\mathcal{A}} && (*) \\
 & \iff \mathcal{A}_{[x \rightarrow u]}(\neg P(x)) = 1 \text{ for some } u \in U^{\mathcal{A}} && \text{(Semantics of } \neg) \\
 & \iff \mathcal{A}(\exists x \neg P(x)) = 1. && \text{(Semantics of } \exists)
 \end{aligned} \tag{2}$$

- (*) If for all assignments of the variable x it holds that $\mathcal{A}_{[x \rightarrow u]}(P(x)) = 1$, then by the semantics of $\forall$ we get $\mathcal{A}(\forall x P(x)) = 1$, a contradiction.